

# Token Distribution Plan

DeepSafe Team

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## 1 Tokenomics Model

### 1.1 Token Allocation

$\$DEF$  is the unique token within the DeepSafe ecosystem, serving as the primary driver for the development of DeepSafe. Within the network,  $\$DEF$  is primarily utilized for gas, governance, and staking, facilitating its rational circulation within the ecosystem. Each role within the network utilizing  $\$DEF$  corresponds to specific rights and incentives.

The maximum supply of  $\$DEF$  is capped at 1 billion, with an initial supply of 350 million. The remaining portion earmarked for issuance is primarily dedicated to long-term incentives for data nodes and validators within the network, fostering the stable and sustainable development of DeepSafe. The specific token distribution details are illustrated in Figure 1:

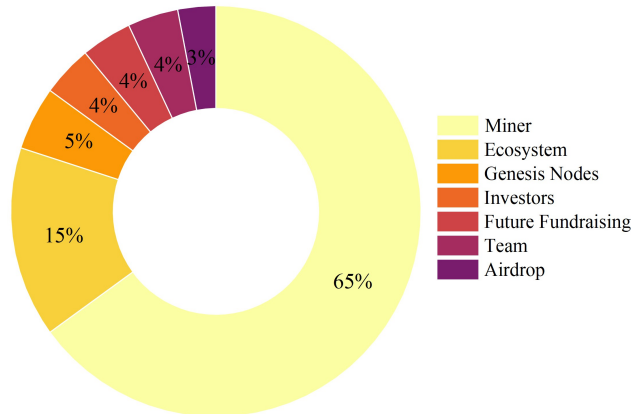


Figure 1:  $\$DEF$  token allocation.

- 65% allocation is earmarked for incentivizing nodes, which will be gradually released over 50 to 55 years.
- 15% allocation is distributed to accelerate the development of the complete DeepSafe ecosystem. It supports pertinent tools and applications within the ecosystem and fosters developer communities.
- Staked tokens for Genesis nodes will be locked for the first year and will follow a 12-month linear release.
- The tokens assigned to early investors will follow a 5% unlock at TGE, with a 3-month cliff, and undergo a 21-month linear release.
- 4% allocation is reserved for future fundraising.

- The team allocation starts with a 36-month cliff and follows a 24-month linear vesting.
- Airdrop allocations are aimed at incentivizing members of the DeepSafe community and are equitably distributed based on the contributions made by community members to the network's development.

## 1.2 Metrics and Functions

DeepSafe's maximum inflationary supply is capped at 650 million within an inflation model defined by upper and lower limits. This model is intricately linked to temporal dynamics and the aggregate staked amount, both critical factors governing the precise estimation of potential risks related to \$DEF\$ supply growth.

### 1.2.1 Key definitions

- **Epoch:** In DeepSafe, the release of node rewards and the modification of parameters operate on an Epoch-based system, with Epoch being the smallest unit of time <sup>1</sup>. In the following context,  $i$  will be used to represent the  $i_{th}$  Epoch since the launch of the mainnet.
- **Number of epoch per year  $N$ :**  $N$  represents the total number of epochs per year.
- **Total supply  $M_i$ :**  $M_i$  represents the total amount of released \$DEF\$ in the market at the beginning of the  $i_{th}$  Epoch.
- **Staking rate  $S_i$ :**  $S_i$  represents the current staking rate of \$DEF\$ at the beginning of the  $i_{th}$  Epoch, calculated as follows:

$$S_i = \frac{\$DEF_{staking}}{M_i} \quad (1)$$

- **Inflation rate  $I_i$ :**  $I_i$  represents the inflation rate of the  $i_{th}$  Epoch.
- **Node reward  $R_i$ :**  $R_i$  represents the total node rewards of the  $i_{th}$  Epoch, released evenly with each block, calculated as follows:

$$R_i = I_i \cdot M_i \quad (2)$$

### 1.2.2 Invariants

- **Initial token supply:**  $M_0 = 3.5 \cdot 10^8$
- **Initial staking rate:**  $S_0 = S_1 = 0.1$
- **Initial inflation rate:**  $I_0 = 0.06$
- **Ideal inflation rate <sup>2</sup>:**  $I_{ideal} = 0.025$

### 1.2.3 Core functions

1. We assume that the value of  $I_i$  is directly influenced by  $S_i$ , aiming to regulate the smooth release of node rewards within the network. It automatically adjusts before the start of each Epoch, represented by the following formula:

$$I_i = \zeta \cdot \max \left\{ I_0 - 0.0048 \cdot \left\lfloor \frac{i}{N} \right\rfloor - 0.027 \cdot \max\{S_i - S_0, 0\}, I_{ideal} \right\} \quad (3)$$

where  $\zeta \in (0, 2]$ , and the initial value of  $\zeta$  is set to 0.75. Further adjustments to  $\zeta$  will be determined through a governance process to ensure system stability.

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<sup>1</sup>Post-mainnet activation, each epoch will be defined as a fixed number of blocks.

<sup>2</sup> $I_{ideal}$  represents the ideal value of the inflation rate.

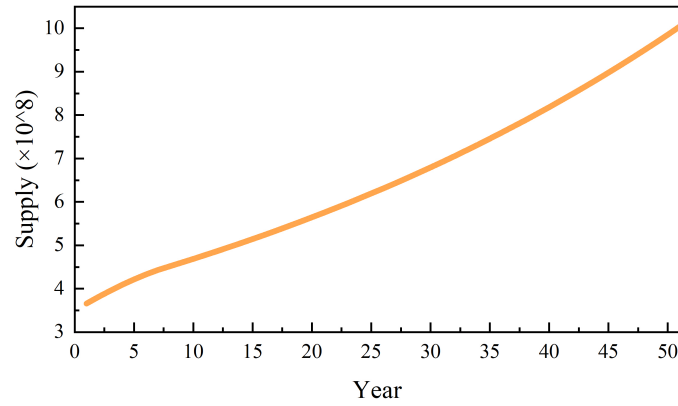
2. The rate of return on node staking represents the percentage gain or profit earned on the amount  $\$DEF$  staked by a node operator within a given period. It quantifies the yield generated from staking activities and reflects the profitability of participation in the network's consensus mechanism. It is derived as follows:

$$RoR_i = \frac{I_i}{S_i} \quad (4)$$

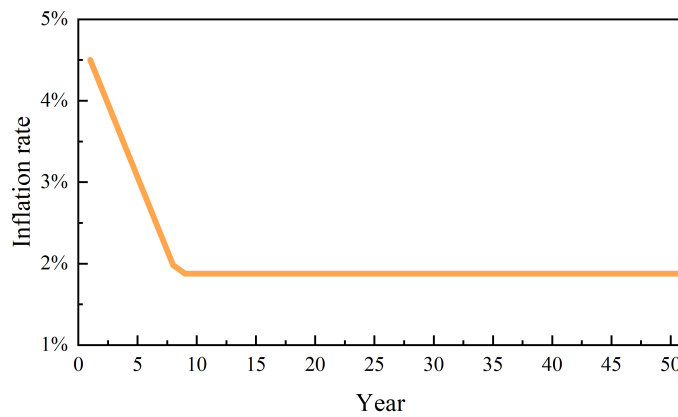
### 1.3 Performance Visualization

Based on the metrics presented in the previous section, we will describe the fluctuation in the  $\$DEF$  token's inflation rate  $I_i$  under specific conditions and the variations in its future supply  $M_i$ . Furthermore, we'll demonstrate the changes in earnings for eligible users engaged in DeepSafe. The issuance of  $\$DEF$  is designated to reward verification nodes and data layer participants, including all delegates. Delegates receive an annualized return based on the  $\$DEF$  token's inflation rate, while specific commissions from delegate earnings are accessible to various nodes in the network.

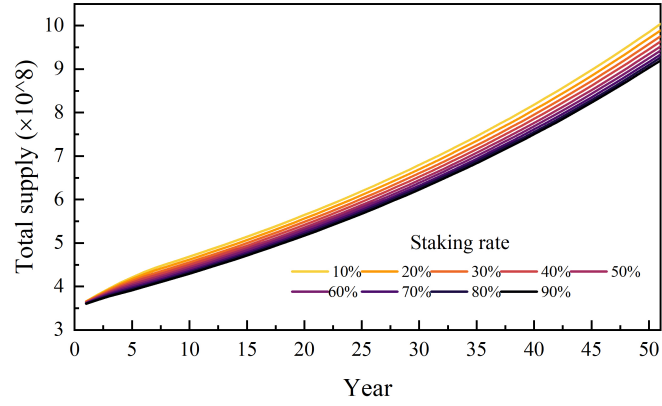
1. Curve depicting the annual supply variation of  $\$DEF$  when staking rate is  $\leq 10\%$ :



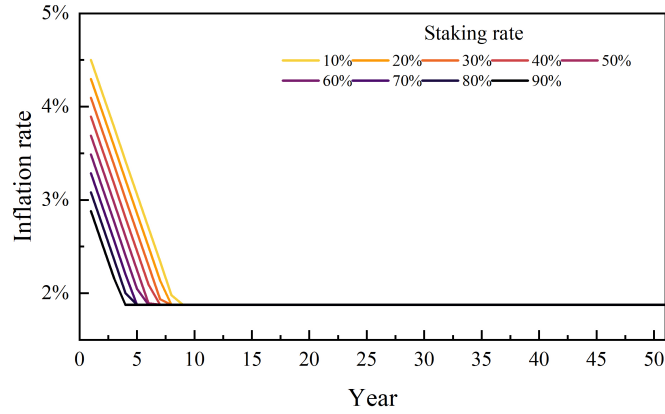
2. Curve showing the annual inflation rate variation of  $\$DEF$  when staking rate is  $\leq 10\%$ :



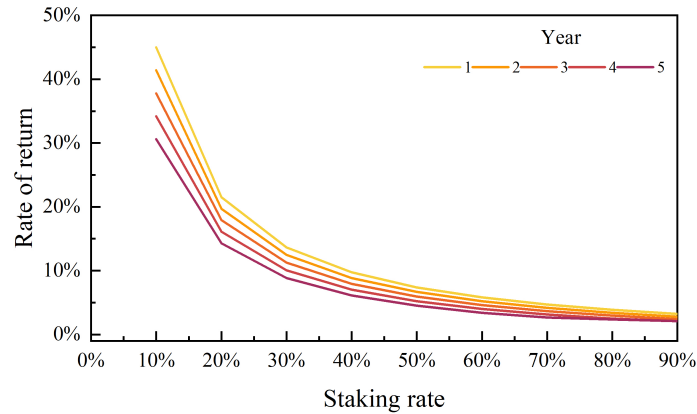
3. Curves displaying the annual supply variation of  $\$DEF$  at different staking rates:



4. Curves illustrating the annual inflation rate variation of  $\$DEF$  at different staking rates:



5. Curves exhibiting the annualized return rate of  $\$DEF$  at different staking rates:



## 1.4 Gas Fee Pool

In DeepSafe, users are required to pay fees while using the network, known as gas. The gas mechanism is dedicated to defining the costs incurred within the CRVA protocol and the data layer and does not encompass any third-party expenses incurred during user interactions.

To ensure the swift fulfillment of user needs within DeepSafe, we have established a more flexible gas mechanism. Presently, you can utilize any widely recognized cryptocurrency to cover the gas fees associated with using DeepSafe. The fee determination will be conducted via community governance, where the gas fees for DeepSafe will be established through  $\$DEF$  voting within each epoch cycle. Billing units will be equivalently calculated based on USDT, considering the equivalence of other tokens to USDT.

Furthermore, payments can also be made using  $\$DEF$ , employing a specific billing algorithm within the DeepSafe protocol layer. There exists a negotiable space between the billing mechanisms of  $\$DEF$  and USDT within the DeepSafe protocol layer, constituting a bargaining margin allocated as negotiating rewards for qualifying users and validators within DeepSafe.

Within DeepSafe, a  $\$DEF$  gas pool will be established. Users voluntarily contribute and burn  $\$DEF$  tokens to this pool, which will be utilized for paying the gas within DeepSafe's CRVA protocol and data layer. The gas within the CRVA protocol layer will remunerate the computational costs of DeepSafe validators, while 50% of the data layer gas will compensate for the computational and storage expenses of DeepSafe full nodes, with the remaining 50% being burnt. The establishment of the gas pool ensures authorization for protocol and data layer payments using other tokens. The USDT payments made by users will be exchanged at an equivalent rate with other tokens based on Oracle quotations. Therefore, transaction fees are not confined to USDT alone and all transaction fees will enter the contract pool, incentivizing all users to contribute to the  $\$DEF$  gas pool.